Notes: Santos and Winton (JF, 2008)

Bank Loans, Bonds, and Information Monopolies across the Business Cycle

Contents

1 Big picture

- **Question.** Do banks use private information acquired through lending relationships to extract higher loan spreads from borrowers, especially when the borrower is risky and outside funding is hard to obtain?
- **Core theory.** The paper tests the Rajan-style informational hold-up idea: banks monitor borrowers and learn private information, but that same information advantage makes it costly for borrowers to switch lenders.
- **Why the business cycle matters.** Hold-up power should be stronger in recessions because borrowers are riskier and outside investors are more concerned about adverse selection. If a borrower cannot credibly access public bond markets, its bank can charge more.
- **Main comparison.** The paper compares bank loan spreads for bank-dependent firms with loan spreads for firms that have access to public debt markets.
- **Identification logic.** If the same bank loan market prices similar observable risk differently for firms with versus without public bond access, and if the difference becomes larger in recessions, this supports an informational monopoly channel.
- **Main data.** The paper uses Loan Pricing Corporation Dealscan loan data linked to Compustat and bond market histories over 1987–2002.
- **Key dependent variable.** The main outcome is *loan spread over LIBOR at origination*, denoted `LOANSPREAD`.
- **Key treatment variable.** The main measure of bond market access is `MRPBOND`, an indicator equal to one if the borrower's most recent bond issue before the loan was a public bond.
- **Main result.** Firms with public bond market access pay lower loan spreads. Loan spreads rise in recessions, but they rise much less for firms with public bond market access.
- **Economic magnitude.** In baseline specifications, bond market access reduces loan spreads by tens of basis points, and recessions raise spreads primarily for bank-dependent firms.
- **Takeaway.** Bank relationships provide monitoring benefits, but they also create state-dependent hold-up costs. Those costs become most visible in recessions.

2 Incremental contribution of the paper

2.1 Contribution relative to relationship-banking theory

- Theoretical work by Diamond, Sharpe, and Rajan emphasizes that banks can create private information and then use it to hold up borrowers.
- Earlier empirical work often studies the number of bank relationships or the choice between bank and public debt.
- This paper directly studies the *price* of bank loans and asks whether relationship-based information monopolies show up in loan spreads.

Interpretation: The paper moves the evidence from relationship structure to contract prices. This is a more direct test of whether informational monopoly power is costly for borrowers.

2.2 Contribution relative to bank-loan pricing research

- Prior bank-loan studies document that loan spreads vary with borrower risk, collateral, loan type, maturity, and syndicate structure.

- Santos and Winton add a business-cycle dimension: the same observable borrower and loan characteristics may be priced differently in recessions.
- They compare the recession sensitivity of spreads for bank-dependent borrowers with the recession sensitivity for firms that can credibly issue public bonds.

Interpretation: The key incremental idea is that informational hold-up is not constant. It should intensify when borrower risk and adverse-selection concerns rise.

2.3 Contribution relative to capital-structure and debt-choice research

- Capital-structure research often asks whether firms use bank debt or public bonds.
- This paper shows that public bond access matters even when firms borrow from banks.
- Public bond access gives borrowers outside options, which limits the bank's pricing power.

Interpretation: Public debt markets discipline banks. Even if the borrower ultimately takes a bank loan, the existence of an outside public-debt option changes the bargaining problem.

2.4 Contribution relative to macro-finance evidence

- The paper uses recessions as a stress test for relationship banking.
- It shows that the cost of bank dependence is largest precisely when external finance is most valuable.
- This connects corporate debt contracting to macroeconomic credit conditions.

Interpretation: The paper's broader message is that relationship banking has cyclical costs: the bank's informational advantage is especially expensive in bad times.

3 Data and measurement

3.1 Loan data and sample construction

- The loan data come from Loan Pricing Corporation's Dealscan database.
- The main sample covers loans originated from 1987 through 2002.
- Dealscan is matched to Compustat to obtain firm accounting variables.
- Because Dealscan focuses on relatively large loans and the Compustat match requires public equity data, the sample is tilted toward larger firms.
- This makes the test conservative: if large public firms still face hold-up costs, smaller private firms likely face even larger costs.

Interpretation: The sample is not the set of all bank loans. It is a sample of relatively large bank loans to firms with publicly observable accounting data, which biases against finding strong informational monopoly effects.

3.2 Loan spread

The main dependent variable is

$$LOANSPREAD_{i\ell t} = \text{loan spread over LIBOR at origination,} \tag{1}$$

where i indexes borrower, ℓ indexes loan, and t indexes origination time.

Interpretation: The spread at origination is the relevant price because it reflects the bank's assessment of borrower risk, outside options, and bargaining power at the time the loan contract is signed.

3.3 Bond market access

The paper uses three related access measures:

- `BOND`: equal to one if the firm issued any bond before the loan.
- `PBOND`: equal to one if the firm issued a public bond before the loan.
- `MRPBOND`: equal to one if the most recent bond before the loan was a public bond.

The paper's preferred access variable is `MRPBOND`. It captures whether the firm appears to have recent access to the public bond market.

Interpretation: A firm that issued a public bond long ago may no longer have meaningful public market access. The most recent bond issue is therefore a sharper proxy for the borrower's current outside option.

3.4 Bank dependence

A firm is treated as bank-dependent if its most recent bond before the loan was not a public bond. Thus bank dependence is the absence of recent public bond market access.

Interpretation: The bank-dependent group includes firms relying primarily on private finance. These are the borrowers for which a bank's private information should be most valuable and most exploitable.

3.5 Recession variable

The recession dummy `REC` equals one for loans originated during recession periods identified using the Stock-Watson Experimental Coincident Recession Index.

Interpretation: Recessions are used to test whether hold-up power rises when borrower risk and adverse selection are higher.

3.6 Firm controls

The paper controls for firm risk and quality using variables such as:

- age;
- sales;
- tangibles;
- sales growth;
- advertising and R&D;
- market-to-book;
- profit margin;
- interest coverage;
- net working capital;
- leverage;

- Altman-style Z-score;
- stock volatility;
- stock returns.

Interpretation: These controls address the possibility that bank-dependent firms simply become riskier in recessions.

3.7 Loan controls

The paper controls for loan characteristics including:

- loan purpose;
- corporate purpose, refinance, takeover, and working capital dummies;
- credit lines and term loans;
- bridge loans;
- loan amount;
- dividend restrictions;
- seniority and collateral;
- renewal status;
- guarantor and sponsor indicators;
- number of lenders;
- single-lender indicator;
- maturity.

Interpretation: The spread results are not merely differences in loan type or syndicate structure. The key effects remain after extensive loan-contract controls.

4 Theory and testable predictions

4.1 Informational hold-up

A bank that monitors a firm learns private information about the firm. If the borrower tries to switch lenders, outside lenders worry that the incumbent bank has bad private information. This creates a lemons problem.

$$\text{Private information from monitoring} \quad \Rightarrow \quad \text{bank bargaining power.} \tag{2}$$

Interpretation: The same information that makes bank monitoring valuable also gives banks pricing power over relationship borrowers.

4.2 Why risk matters

The hold-up problem is stronger when the borrower is riskier. Riskier borrowers are harder for outside lenders to evaluate, so outside lenders discount the possibility that a borrower leaving its bank is a lemon.

$$\frac{\partial \text{hold-up power}}{\partial \text{borrower risk}} > 0. \tag{3}$$

Interpretation: Recessions are a natural state in which borrower risk rises. Thus hold-up should be more visible during recessions.

4.3 Hypothesis 1: bond access lowers spreads

Firms with public bond access have a stronger outside option and should pay lower bank loan spreads:

$$\beta_{Access} < 0. \tag{4}$$

Interpretation: Public market access disciplines banks even when the firm chooses to borrow from a bank.

4.4 Hypothesis 2: recessions raise spreads

Loan spreads should rise in recessions because borrower risk and lender risk premia rise:

$$\beta_{REC} > 0. \tag{5}$$

Interpretation: This is the basic macro credit-spread prediction.

4.5 Hypothesis 3: recession spread increases are larger for bank-dependent borrowers

If informational hold-up matters, recessions should increase spreads more for borrowers without public bond access:

$$\beta_{REC\times Access} < 0. \tag{6}$$

Interpretation: The negative interaction is the paper's signature test. It says that access to public debt markets dampens the recession-induced increase in bank loan spreads.

5 Empirical specifications

5.1 Baseline spread regression

The main regression is:

$$LOANSPREAD_{i\ell t} = \alpha + \beta_1 Access_{it} + \beta_2 REC_t + \beta_3 (REC_t \times Access_{it}) + X_{it}'\Gamma + Z_{i\ell t}'\Delta + \eta_s + \varepsilon_{i\ell t}, \tag{7}$$

where X_{it} are firm controls, $Z_{i\ell t}$ are loan controls, and η_s are sector fixed effects.

Interpretation: The test is whether $\beta_1 < 0$, $\beta_2 > 0$, and $\beta_3 < 0$.

5.2 Preferred access specification

The preferred version uses `MRPBOND`:

$$LOANSPREAD = \alpha + \beta_1 MRPBOND + \beta_2 REC + \beta_3 REC \times MRPBOND + Controls + \varepsilon. \tag{8}$$

Interpretation: A negative β_3 means that firms with recent public bond access are insulated from the recession increase in bank loan spreads.

5.3 Risk-interaction specification

The paper further interacts access with firm-risk proxies:

$$LOANSPREAD = \alpha + \beta MRPBOND + \theta RiskProxy + \phi(MRPBOND \times RiskProxy) + Controls + \varepsilon. \tag{9}$$

Interpretation: If the informational monopoly effect is stronger for riskier firms, access to public bond markets should reduce spreads more when firm risk is high.

5.4 Instrumental variables specification

The paper instruments for recent public bond market access using variables related to exogenous access to public debt markets:

$$MRPBOND_{it} = \Phi(W_{it}\Pi) + u_{it}, \tag{10}$$

where W includes instruments such as exchange listing, inclusion in major indices, Lehman bond index eligibility, and industry-level public bond market access.

The fitted value is then used in the spread regression:

$$LOANSPREAD = \alpha + \beta_1 \widehat{MRPBOND} + \beta_2 REC + \beta_3 REC \times \widehat{MRPBOND} + Controls + \varepsilon. \tag{11}$$

Interpretation: The IV approach addresses the concern that public bond access is endogenous to unobserved borrower quality.

6 Identification and empirical credibility

6.1 Key identification challenge

The main challenge is that public bond market access is not randomly assigned. Firms with bond access are larger, safer, older, and less information opaque. They may also be less affected by recessions for reasons unrelated to bank hold-up.

Interpretation: The paper must show that the recession interaction is not simply a risk-composition effect.

6.2 How the paper addresses borrower risk

- It controls for many firm-level risk proxies, including leverage, interest coverage, Z-score, stock volatility, and stock returns.
- It controls for loan-level contract terms and syndicate structure.
- It runs difference-in-differences tests on observable risk proxies.
- It uses IV specifications for bond market access.

Interpretation: No observational design can fully prove exogeneity, but the robustness battery is aimed directly at selection on risk.

6.3 Why the recession interaction is informative

A simple level difference in spreads between bond-access firms and bank-dependent firms could reflect permanent credit-quality differences. The recession interaction is harder to explain with static differences alone.

$$\text{Key evidence} = \left[Spread_{Rec}^{NoAccess} - Spread_{Exp}^{NoAccess}\right] - \left[Spread_{Rec}^{Access} - Spread_{Exp}^{Access}\right]. \tag{12}$$

Interpretation: The paper's key comparison is a difference-in-differences across borrower access status and business-cycle state.

6.4 Limitations

- The sample contains relatively large borrowers in Dealscan and Compustat, not small private firms.
- Public bond access remains partly endogenous despite IV and controls.
- The loan spread at origination captures pricing but not all nonprice terms or renegotiation margins.
- The recession periods are few, so the time-series variation is limited.

Interpretation: These limitations make the results conservative but also mean the paper is strongest as evidence of pricing patterns consistent with bank information monopolies.

7 Tables and figures

7.1 Table I and Table II: variables and sample characteristics

Read:

- Table I defines all variables used in the paper, including loan spreads, firm controls, loan controls, bond access measures, and recession indicators.
- Table II compares bank-dependent borrowers with non-bank-dependent borrowers.
- Bank-dependent firms pay much higher average loan spreads: about 226 basis points versus 125 basis points.
- Bank-dependent firms are younger, smaller, less profitable, have lower Z-scores, and are less likely to borrow from nonrelationship banks.

Economic message: The sample differences show why extensive risk controls are necessary. Bank-dependent firms look riskier, but the paper argues that observable risk does not explain the full recession interaction.

Table I
Variable Definitions

Variable	Definition
LOANSPREAD	Loan spread over LIBOR at the time of the loan origination.
AGE	Age in years.
OLDERFIRMS	Dummy variable equal to one for firms older than 20 years.
SALES	Sales in hundreds of millions of dollars at 1980 prices.
SALESGROWTH	The firm's sales growth in the year prior to the loan.
TANGIBLES	Property, plant, and equipment plus inventories over assets.
TANGIBLES-B50	Dummy variable equal to one for firms with TANGIBLES below the sample median.
ADVERTISING	Advertising expenses over sales.
R&D	Research and development expenses over sales.
MKTBOOK	Market-to-book ratio.
PROFMARGIN	Net income over sales.
INTCOVERAGE	Earnings before taxes and depreciation over interest expenses. When the firm has no interest expenses this variable is set equal to earnings before taxes and depreciation.
INTCOVERAGE-B50	Dummy variable equal to one for firms with INTCOVERAGE below the sample median.
NWCDEBT	Current assets minus current liabilities over total debt. When the firm has no debt this variable is set equal to current assets minus current liabilities.
LEVERAGE	Total debt over assets.
LEVERAGE-A50	Dummy variable equal to one for firms with LEVERAGE above the sample median.
ZSCORE	Computed with quarterly data over the previous 3 years. See footnote (9) for the definition of the ZSCORE that we use.
ZSCORE-B50	Dummy variable equal to one for firms with a ZSCORE below the sample median.
STOCKVOL	Standard deviation of stock return over the year leading up to the loan date.
STOCKVOL-A50	Dummy variable equal to one for firms with STOCKVOL above the sample median.
STOCKRET	Return on the company's stock over the preceeding year minus the return on the equally weighted CRSP index of NYSE/AMEX stocks over the year.
RLENDINGnarrow	Dummy variable equal to one if the firm borrowed at least once from the lead underwriter(s) in the loan syndicate over the previous 3 years.
RLENDINGbroad	Dummy variable equal to one if the firm borrowed at least once from any bank in the loan syndicate over the previous 3 years.
CORPURPOSES	Dummy variable equal to one when loan is for corporate purposes.
REFINANCE	Dummy variable equal to one when loan is to repay existing debt.
TAKEOVER	Dummy variable equal to one when loan is for takeover purposes.
WORKCAPITAL	Dummy variable equal to one when loan is for working capital purposes.
CREDITLINE	Dummy variable equal to one for lines of credit.
TERMLOAN	Dummy variable equal to one for term loans.
BRIDGELOAN	Dummy variable equal to one for bridge loans.
LOANAMT	Loan amount in hundreds of millions of dollars at 1980 prices.
DIVRESTRICT	Dummy variable equal to one when borrower is subject to dividend restrictions.

(*continued*)

Table I, part 1: variable definitions

ıriable	Definition
ENIOR	Dummy variable equal to one when loan is senior.
ECURED	Dummy variable equal to one when loan is secured.
ENEWAL	Dummy variable equal to one when loan is a renewal.
UARANTOR	Dummy variable equal to one when borrower has a guarantor.
PONSOR	Dummy variable equal to one when borrower has a sponsor.
ENDERS	Number of lenders in the loan syndicate.
NGLENDER	Dummy variable equal to one when the loan syndicate has a single lender.
ATURITY	Maturity of the loan in years.
OND	Dummy variable equal to one for firms that issued bonds prior to their loan.
BOND	Dummy variable equal to one for firms that issued public bonds prior to the loan.
BONDblg	Dummy variable equal to one for those firms with PBOND=1 whose last public bond prior to the loan was rated below grade.
BONDnrt	Dummy variable equal to one for those firms with PBOND=1 whose last public bond prior to the loan was not rated.
RPBOND	Dummy variable equal to one for firms whose most recent bond prior to the loan was a public bond.
RPBONDblg	Dummy variable equal to one for firms whose last bond prior to the loan was a public bond with a below grade rating.
RPBONDnrt	Dummy variable equal to one for firms whose last bond prior to the loan was a public bond not rated.
EC	Dummy variable equal to one for loans borrowed during a recession as defined by the Stock–Watson Experimental Coincident Recession Index. See Section II.A for the dates of the recessions during our sample period.
YSE	Dummy variable equal to one for firms that trade in the NYSE.
EHMAN	Dummy variable equal to one for firms that issued bonds in amounts larger than the minimum necessary to be included in the Lehman Brothers' Corporate Bond Index.
&P500	Dummy variable equal to one for firms that are members of the S&P500 index.
IDACCESSmrpbond	Percentage of the firms in the same industry (as defined by the two-digit SIC code) that have access to the bond market as defined by MRPBOND.
IDACCESSpbond	Percentage of the firms in the same industry (as defined by the two-digit SIC code) that have access to the bond market as defined by PBOND.
SPREAD	Treasury yield curve slope computed as the difference between the 30- and 5-year Treasury bonds.
BBSPREAD	Spread of new BBB rated bonds over new AAA rated bonds.

ırposes, to fund takeovers, and to refinance; they are less likely to borrow ırough credit lines and to use bridge loans.

Figure 1 graphs the numbers of loans to bank-dependent firms and to firms ith access to the bond market over time. As can be seen, there are many ınes more loans to bank-dependent borrowers than to non-bank-dependent

Table I, part 2: variable definitions

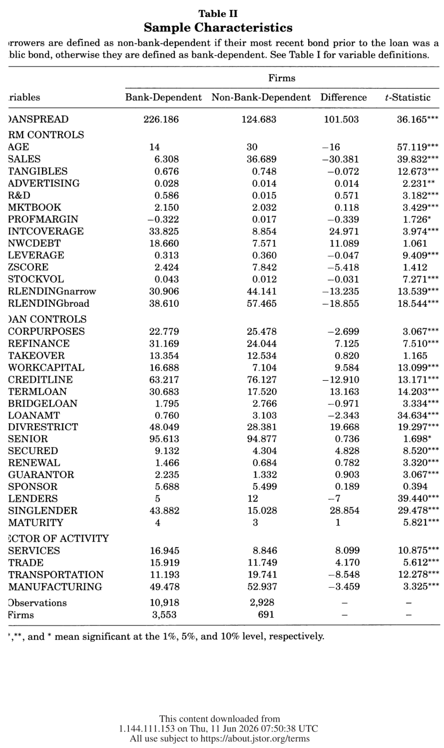
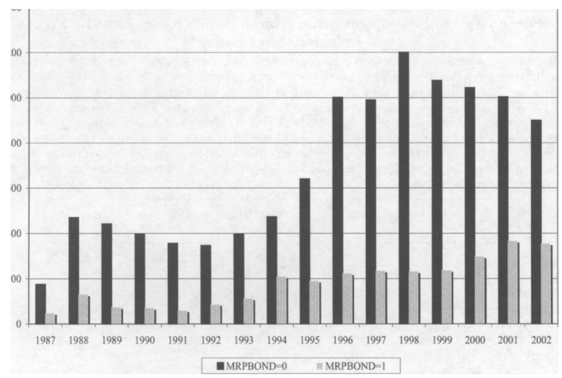


Table II
Sample Characteristics

rrowers are defined as non-bank-dependent if their most recent bond prior to the loan was a
blic bond, otherwise they are defined as bank-dependent. See Table I for variable definitions.

riables	Firms			
	Bank-Dependent	Non-Bank-Dependent	Difference	t-Statistic
)ANSPREAD	226.186	124.683	101.503	36.165***
RM CONTROLS				
AGE	14	30	−16	57.119***
SALES	6.308	36.689	−30.381	39.832***
TANGIBLES	0.676	0.748	−0.072	12.673***
ADVERTISING	0.028	0.014	0.014	2.231**
R&D	0.586	0.015	0.571	3.182***
MKTBOOK	2.150	2.032	0.118	3.429***
PROFMARGIN	−0.322	0.017	−0.339	1.726*
INTCOVERAGE	33.825	8.854	24.971	3.974***
NWCDEBT	18.660	7.571	11.089	1.061
LEVERAGE	0.313	0.360	−0.047	9.409***
ZSCORE	2.424	7.842	−5.418	1.412
STOCKVOL	0.043	0.012	−0.031	7.271***
RLENDINGnarrow	30.906	44.141	−13.235	13.539***
RLENDINGbroad	38.610	57.465	−18.855	18.544***
)AN CONTROLS				
CORPURPOSES	22.779	25.478	−2.699	3.067***
REFINANCE	31.169	24.044	7.125	7.510***
TAKEOVER	13.354	12.534	0.820	1.165
WORKCAPITAL	16.688	7.104	9.584	13.099***
CREDITLINE	63.217	76.127	−12.910	13.171***
TERMLOAN	30.683	17.520	13.163	14.203***
BRIDGELOAN	1.795	2.766	−0.971	3.334***
LOANAMT	0.760	3.103	−2.343	34.634***
DIVRESTRICT	48.049	28.381	19.668	19.297***
SENIOR	95.613	94.877	0.736	1.698*
SECURED	9.132	4.304	4.828	8.520***
RENEWAL	1.466	0.684	0.782	3.320***
GUARANTOR	2.235	1.332	0.903	3.067***
SPONSOR	5.688	5.499	0.189	0.394
LENDERS	5	12	−7	39.440***
SINGLENDER	43.882	15.028	28.854	29.478***
MATURITY	4	3	1	5.821***
ECTOR OF ACTIVITY				
SERVICES	16.945	8.846	8.099	10.875***
TRADE	15.919	11.749	4.170	5.612***
TRANSPORTATION	11.193	19.741	−8.548	12.278***
MANUFACTURING	49.478	52.937	−3.459	3.325***
Observations	10,918	2,928	–	–
Firms	3,553	691	–	–

*,**, and * mean significant at the 1%, 5%, and 10% level, respectively.



igure 1. Time distribution of loans of bank-dependent and non-bank-dependent firms.

orrowers, and loans to bank-dependent borrowers seem to be more sensitive
credit conditions. For example, although both types of loans decrease in 1989
he year of our "mini recession"), loans to bank-dependent firms continue to
ecline in the early 1990s whereas those to non-bank-dependent firms are rel-
tively stable. A similar pattern occurs in the late 1990s, where loans to bank-
ependent firms peak in 1998 (the year of the Russian and Long Term Capital

Table II: sample characteristics *Figure 1: loan counts over time*

7.2 Figure 1 and Table III: bond access and relationship switching

Read:

- Figure 1 shows the time distribution of loans to bank-dependent and non-bank-dependent firms.
- Loans to bank-dependent firms are much more numerous and more cyclical.
- Table III shows that bank-dependent firms are less likely to borrow from a different bank than their relationship lender.
- The switching gap exists in both expansions and recessions.

Economic message: Bank-dependent firms appear more tied to relationship lenders, consistent with the information-monopoly channel.

Table III
Percentage of Loans Borrowed from a Bank Other than the Firm's Relationship Lender

rcentage of loans borrowed from a bank other than the firm's relationship lender. A bank is fined as a relationship lender if it acted as a lead underwriter at least once for the loan(s) the m took out in the 3 years prior to the current loan (this definition matches our definition of ENDINGnarrow). Percentages are computed after we exclude the first loan of each firm in the mple given that we do not have information on the firm's lending relationships prior to that loan. rrowers are defined as non-bank-dependent if their most recent bond prior to the loan was a blic bond, otherwise they are defined as bank dependent. Recessions are defined by the Stock–atson Experimental Coincident Recession Index. See Section II.A for the dates of the recessions ring our sample period.

	Bank-Dependent-Firms	Non-Bank-Dependent Firms	Difference	t-Statistic
		All loans		
	43.5	53.0	9.5	6.38
		Depending on the state of the economy		
pansions	43.8	53.0	9.2	6.33
cessions	42.3	53.3	11.0	3.57

urce: Authors' computations.

ore likely to borrow from a relationship bank, which is what our model would edict.
In Table III, we also report the percentages of firms that switch banks expansions versus recessions. Once again, we classify firms as non-bank-pendent if their most recent bond issue was public. Although bank-dependent

Table III: relationship-lender switching

Table IV
Loan Spreads at Issue Date for Bank-Dependent and Non-Bank-Dependent Borrowers

e dependent variable is LOANSPREAD. See Table I for variable definitions. Models estimated th robust standard errors and clustered by firm to correct for correlation across observations of given firm. Values in parentheses are p-values.

riables	1	2	3	4	5	6	7
NSTANT	237.990	230.747	226.186	200.831	233.190	224.845	220.722
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ND	−74.537				−74.386		
	(0.000)				(0.000)		
OND		−90.244				−85.745	
		(0.000)				(0.000)	
RPBOND			−101.503				−95.263
			(0.000)				(0.000)
C				19.523	26.247	31.079	27.797
				(0.000)	(0.000)	(0.000)	(0.000)
C · BOND					−5.076		
					(0.545)		
C · PBOND						−24.771	
						(0.005)	
C · MRPBOND							−31.507
							(0.000)
ljusted-R^2	0.0690	0.0840	0.0863	0.31	0.0736	0.0894	0.0912
bservations	13,846	13,846	13,846	13,846	13,846	13,846	13,846

y bond typically pay 75 basis points less than other firms, firms that have sued public bonds pay 90 basis points less than other firms, and firms whose ost recent bond was publicly issued pay 102 basis points less than other firms. l of these results are highly significant and consistent with Hypothesis 1. e larger effect for public bonds is consistent with our prediction that issuers

Table IV: univariate spread regressions

7.3 Table IV: univariate spread regressions

Read:

- Firms with any bond, public bond, or most recent public bond access pay lower loan spreads.
- Recession loans have higher spreads.
- The interaction between recession and bond access is negative.
- The effect is strongest for the most recent public bond measure.

Economic message: The univariate results already match the theory: bond access lowers spreads and dampens recession pricing increases.

7.4 Table V: multivariate analysis

Read:

- Table V adds firm controls and loan controls.
- `MRPBOND` remains strongly negative: recent public bond access lowers spreads.
- `REC` remains positive: recessions increase loan spreads.
- `REC` × `MRPBOND` is negative and significant in the main models.
- The results are robust whether access is measured by `MRPBOND`, `PBOND`, or related measures.

Economic message: The main pricing pattern survives detailed controls. This is the core empirical evidence for state-dependent informational hold-up.

Table V

Loan Spreads for Bank-Dependent and Nondependent Borrowers: Multivariate Analysis

ıe dependent variable is LOANSPREAD. LINTCOVERAGE = log(1 + INTCOVERAGE), with ꞮTCOVERAGE truncated at 0. LTIMRPBOND: Log of 1 plus the number of months since the firm ʂued its most recent public bond. LAGE = Log(AGE). LSALES = LOG(SALES). LLENDERS = ⅁G(LENDERS). LMATURITY = Log(MATURITY). See Table I for variable definitions. Included the regressions but not shown in the table are also dummy variables for the issuer's sector of tivity as defined by one-digit SIC code. Models are estimated with robust standard errors and ıstered by firm to correct for correlation across observations of a given firm. Values in parentheses e p-values.

ıriables	1	2	3	4	5	6
ƆNSTANT	342.673	339.422	628.031	629.020	629.694	630.476
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
RPBOND	−17.410	−36.940	−38.073	−48.649		
	(0.000)	(0.000)	(0.000)	(0.000)		
RPBONDblg		37.206		34.670		
		(0.000)		(0.000)		
RPBONDnrt		44.295		35.484		
		(0.008)		(0.007)		
ƎOND					−18.324	−33.729
					(0.029)	(0.000)
ƎONDblg						30.480
						(0.000)
ƎONDnrt						31.940
						(0.004)
ƐC	32.801	32.278	22.492	21.790	22.067	21.555
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ƐC · MRPBOND	−21.598	−15.797	−20.212	−16.483		
	(0.002)	(0.037)	(0.001)	(0.010)		
ƐC · MRPBONDblg		6.050		8.322		
		(0.663)		(0.448)		
ƐC · MRPBONDnrt		−31.824		−11.709		
		(0.293)		(0.666)		
ƐC · PBOND					−11.301	−15.119
					(0.068)	(0.023)
ƐC · PBONDblg						29.628
						(0.009)
ƐC · PBONDnrt						−23.645
						(0.322)
RM CONTROLS						
LAGE	−5.401	−4.922	−5.832	−5.401	−6.442	−5.777
	(0.008)	(0.016)	(0.002)	(0.003)	(0.001)	(0.002)
LSALES	−44.186	−41.951	−18.100	−16.152	−19.756	−16.946
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ꞮANGIBLES	−30.331	−29.449	−24.483	−24.111	−24.565	−23.463
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ADVERTISING	0.328	1.484	−4.591	−3.487	−5.467	−3.947
	(0.988)	(0.946)	(0.829)	(0.870)	(0.794)	(0.852)
R&D	−0.459	−0.488	−0.135	−0.165	−0.104	−0.148
	(0.570)	(0.549)	(0.864)	(0.835)	(0.894)	(0.850)

(*continued*)

ıriables	1	2	3	4	5	6
MKTBOOK	−5.965	−5.713	−4.566	−4.388	−4.692	−4.425
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
PROFMARGIN	−0.238	−0.242	−0.130	−0.135	−0.130	−0.135
	(0.107)	(0.105)	(0.372)	(0.354)	(0.368)	(0.350)
LINTCOVERAGE	−32.260	−32.180	−27.777	−27.602	−27.593	−27.533
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
NWCDEBT	0.002	0.001	0.003	0.002	0.003	0.003
	(0.327)	(0.355)	(0.030)	(0.032)	(0.024)	(0.028)
LEVERAGE	26.999	23.559	35.245	33.076	34.144	30.389
	(0.002)	(0.006)	(0.000)	(0.000)	(0.000)	(0.001)
ZSCORE	−0.014	−0.014	−0.013	−0.013	−0.013	−0.013
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ƆAN CONTROLS						
LTIMRPBOND			5.027	3.692	3.552	2.758
			(0.000)	(0.007)	(0.008)	(0.034)
CORPURPOSES			−21.801	−22.886	−21.570	−22.756
			(0.000)	(0.000)	(0.000)	(0.000)
REFINANCE			−16.914	−18.050	−16.338	−17.995
			(0.000)	(0.000)	(0.000)	(0.000)
TAKEOVER			3.161	2.154	3.479	2.064
			(0.511)	(0.655)	(0.470)	(0.668)
WORKCAPITAL			−21.380	−22.836	−20.522	−22.230
			(0.000)	(0.000)	(0.000)	(0.000)
CREDITLINE			−21.261	−20.712	−21.255	−20.101
			(0.000)	(0.000)	(0.000)	(0.001)
TERMLOAN			15.787	15.948	16.164	16.613
			(0.011)	(0.009)	(0.010)	(0.008)
BRIDGELOAN			98.385	96.776	99.051	97.905
			(0.000)	(0.000)	(0.000)	(0.000)
LLOANAMT			−21.543	−21.732	−21.712	−21.874
			(0.000)	(0.000)	(0.000)	(0.000)
DIVRESTRICT			17.674	16.865	18.333	16.898
			(0.000)	(0.000)	(0.000)	(0.000)
SENIOR			−16.629	−14.841	−16.423	−15.137
			(0.010)	(0.022)	(0.011)	(0.020)
SECURED			60.307	59.983	60.975	59.352
			(0.000)	(0.000)	(0.000)	(0.000)
RENEWAL			−6.241	−7.066	−5.965	−6.714
			(0.442)	(0.383)	(0.461)	(0.403)
GUARANTOR			15.377	15.402	16.672	15.988
			(0.120)	(0.119)	(0.091)	(0.107)
SPONSOR			52.686	49.984	52.783	50.132
			(0.000)	(0.000)	(0.000)	(0.000)
LLENDERS			−2.872	−3.058	−3.220	−3.109
			(0.232)	(0.203)	(0.187)	(0.201)
SINGLENDER			6.258	6.069	5.767	6.011
			(0.135)	(0.146)	(0.169)	(0.150)
LMATURITY			−18.808	−20.096	−18.275	−19.924
			(0.000)	(0.000)	(0.000)	(0.000)
ɖjusted-R^2	0.3625	0.3658	0.4626	0.4652	0.4609	0.4649
observations	13,846	13,846	13,846	13,846	13,846	13,846

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Table V, part 1: main multivariate results *Table V, part 2: controls and fit*

7.5 Table VI: interactions with firm risk proxies

Read:

- Table VI interacts `MRPBOND` with firm risk proxies such as low tangibles, high leverage, low Z-score, low interest coverage, and high stock volatility.
- The access discount is larger for many riskier borrowers.
- This pattern is consistent with the model's prediction that hold-up is stronger for risky firms.

Economic message: The bond-access advantage is not merely a level effect. It is larger when borrowers are more exposed to risk and adverse selection.

Table VI

Loan Spreads for Bank-Dependent and Nondependent Borrowers: Interaction of Bond Market Access and Firm Risk Measures

e dependent variable is LOANSPREAD. FIRM CONTROLS and LOAN CONTROLS coincide th the set of firm controls and loan controls, respectively, used in Table IV. See Table I for variable finitions. Included in the regressions but not shown in the table are also dummy variables for e issuer's sector of activity as defined by one-digit SIC code. Models are estimated with robust andard errors and clustered by firm to correct for correlation across observations of a given firm. lues in parentheses are p-values.

riables	1	2	3	4	5
ONSTANT	604.140	631.419	622.366	547.534	492.724
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
RPBOND	−42.114	−34.970	−34.331	−23.829	−36.999
	(0.000)	(0.000)	(0.000)	(0.005)	(0.000)
NGIBLES-B50	11.000				
	(0.001)				
RPBOND · TANGIBLES-B50	−0.560				
	(0.927)				
VERAGE-A50		13.704			
		(0.000)			
RPBOND · LEVERAGE-A50		−9.987			
		(0.000)			
CORE-B50			6.905		
			(0.039)		
RPBOND · ZSCORE-B50			−14.156		
			(0.018)		
TCOVERAGE-B50				66.269	
				(0.000)	
RPBOND · INTCOVERAGE-B50				−24.691	
				(0.000)	
OCKVOL-A50					68.304
					(0.000)
RPBOND · STOCKVOL-A50					13.846
					(0.120)
C	18.087	17.801	17.820	17.548	10.195
	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
RM CONTROLS	IN	IN	IN	IN	IN
AN CONTROLS	IN	IN	IN	IN	IN
ljusted-R^2	0.4616	0.4610	0.4624	0.4582	0.5006
bservations	13,846	13,846	13,846	13,846	12141

Again, the results are generally consistent with our predictions. The coefcient on bond market access is always negative and significant, whereas e coefficient for the risk proxy dummy is always positive and significant. e coefficient on the interaction term is negative and significant in three ses (leverage, Z-score, and interest coverage), negative and insignificant in e case (tangible assets), and positive and insignificant in one case (stock

Table VI: interaction of bond access and firm-risk measures

7.6 Tables VII and VIII: instrumental-variable analysis

Read:

- Table VII reports first-stage probit models for `MRPBOND`.
- Instruments include variables capturing public bond market visibility and access, such as exchange listing, S&P500 membership, Lehman bond index eligibility, and industry public debt access.
- Table VIII reports second-stage spread regressions using fitted `MRPBOND`.
- The fitted public-bond-access effect remains negative, and the recession interaction remains consistent with the baseline results.

Economic message: The IV results address the concern that bond market access simply proxies for unobserved borrower quality.

Table VII

Determinants of Firms' Access to the Bond Market: First Stage of a Two-Stage Procedure for Market Access

ıe dependent variable is MRPBOND. See Table 1 for variable definitions. Probit models are timated with robust standard errors and clustered by firm to correct for correlation across ob- rvations of a given firm. Included in the regressions but not shown in the table are also dummy riables for the issuer's sector of activity as defined by one-digit SIC code. Values in parentheses e p-values.

ıriables	1	2	3	4	5	6
ɔNSTANT	−2.390	−2.340	−2.371	−2.365	−2.417	−2.413
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ľSE	0.507	0.506	0.493	0.499	0.494	0.501
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ΞHMAN	0.982	0.981	0.982	0.979	0.982	0.979
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ːP500	0.357	0.354	0.367	0.366	0.370	0.369
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
⅃DERFIRMS		0.067	0.061	0.063		
		(0.552)	(0.589)	(0.580)		
IDACCESSmrpbond			0.976		0.983	
			(0.031)		(0.030)	
IDACCESSpbond				0.457		0.464
				(0.215)		(0.206)
RM CONTROLS						
LAGE	0.194	0.165	0.160	0.163	0.186	0.190
	(0.000)	(0.004)	(0.005)	(0.005)	(0.000)	(0.000)
LSALES	0.232	0.231	0.220	0.223	0.221	0.224
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
TANGIBLES	0.141	0.140	0.054	0.090	0.054	0.090
	(0.264)	(0.266)	(0.669)	(0.476)	(0.668)	(0.476)
ADVERTISING	0.770	0.763	0.703	0.743	0.710	0.750
	(0.206)	(0.208)	(0.252)	(0.222)	(0.249)	(0.219)
R&D	−1.535	−1.502	−1.410	−1.428	−1.439	−1.458
	(0.078)	(0.082)	(0.092)	(0.092)	(0.088)	(0.088)
MKTBOOK	0.034	0.034	0.034	0.034	0.035	0.034
	(0.009)	(0.009)	(0.007)	(0.007)	(0.007)	(0.007)
PROFMARGIN	0.008	0.008	0.008	0.008	0.007	0.008
	(0.489)	(0.496)	(0.463)	(0.482)	(0.452)	(0.472)
LINTCOVERAGE	−0.100	−0.098	−0.096	−0.096	−0.097	−0.097
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
NWCDEBT	−0.000	−0.000	−0.000	−0.000	−0.000	−0.000
	(0.837)	(0.871)	(0.818)	(0.840)	(0.786)	(0.808)
LEVERAGE	0.382	0.387	0.377	0.383	0.372	0.377
	(0.003)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
ZSCORE	0.000	0.000	0.000	0.000	0.000	0.000
	(0.373)	(0.376)	(0.396)	(0.393)	(0.394)	(0.391)
ɛeudo-R^2	0.3854	0.3855	0.3867	0.3859	0.3866	0.3858
value for H1: ($\beta = 0$)	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Observations	13,846	13,846	13,846	13,846	13,846	13,846

Table VII: first-stage public bond access models

ıaɒıe vııı

Loan Spreads for Bank-Dependent and Nondependent Borrowers: Second Stage of a Two-Stage Procedure for Market Access

ıe dependent variable is LOANSPREAD. The models in this table are the second stage of a o-stage procedure for bond market access. For the first stage models, including the additional riables used in each model to determine access to the bond market, see Table VII. $\widehat{MRPBOND}$: tted values for MRPBOND are computed based on models in Table VII. See Table I for variable finitions. Included in the regressions but not shown in the table are also dummy variables for e issuer's sector of activity as defined by one-digit SIC code. Models are estimated with robust andard errors and clustered by firm to correct for correlation across observations of a given firm. ılues in parentheses are p-values.

ıriables	1	2	3	4	5	6
ɔNSTANT	341.540	341.449	341.240	341.385	341.322	341.469
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
$\widehat{RPBOND}$	−33.059	−34.207	−35.996	−34.437	−34.953	−33.364
	(0.010)	(0.007)	(0.005)	(0.008)	(0.007)	(0.010)
ΞC	32.230	32.213	32.411	32.363	32.427	32.380
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ΞC · $\widehat{MRPBOND}$	−21.715	−21.843	−23.095	−22.589	−22.985	−22.479
	(0.052)	(0.050)	(0.040)	(0.044)	(0.041)	(0.046)
RM CONTROLS						
LAGE	−4.624	−4.566	−4.463	−4.546	−4.516	−4.601
	(0.028)	(0.030)	(0.034)	(0.031)	(0.032)	(0.029)
LSALES	−41.829	−41.653	−41.346	−41.596	−41.506	−41.760
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
TANGIBLES	−30.054	−30.034	−30.005	−30.032	−30.023	−30.050
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ADVERTISING	1.213	1.279	1.387	1.296	1.327	1.235
	(0.956)	(0.954)	(0.950)	(0.953)	(0.952)	(0.956)
R&D	−0.487	−0.490	−0.493	−0.490	−0.491	−0.488
	(0.552)	(0.550)	(0.548)	(0.550)	(0.549)	(0.551)
MKTBOOK	−5.828	−5.818	−5.799	−5.814	−5.809	−5.824
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
PROFMARGIN	−0.237	−0.237	−0.237	−0.237	−0.237	−0.237
	(0.111)	(0.111)	(0.112)	(0.111)	(0.111)	(0.111)
LINTCOVERAGE	−32.499	−32.516	−32.542	−32.519	−32.526	−32.503
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
NWCDEBT	0.002	0.001	0.001	0.001	0.001	0.001
	(0.356)	(0.358)	(0.364)	(0.360)	(0.362)	(0.358)
LEVERAGE	28.575	28.693	28.895	28.729	28.788	28.619
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
ZSCORE	−0.014	−0.014	−0.014	−0.014	−0.014	−0.014
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
ljusted R^2	0.361	0.361	0.360	0.361	0.361	0.361
ɔbservations	13,846	13,846	13,846	13,846	13,846	13,846

gnificant impacts makes this unlikely. Furthermore, although inclusion in ıese indexes increases visibility, it is difficult to see why it would have any rect effect on risk other than by making access to financial markets easier, hich is the point we want to focus on. The only variable that might proxy for

Table VIII: IV loan spread regressions

7.7 Tables IX and X: robustness checks

Read:

- Table IX adds controls for alternative recession definitions, stock market performance, and borrower risk measures.
- The main access and recession interaction results remain robust.
- Table X investigates relationship lending and syndicate structure.
- Relationship-lending controls do not eliminate the main results.

Economic message: The results are not driven by a particular recession definition, stock-market condition, or relationship-lending proxy.

Table IX

Loan Spreads for Bank-Dependent and Nondependent Borrowers: Robustness Checks: Additional Risk Controls

The dependent variable is LOANSPREAD. Models 1 and 2 control additionally for SALESGROWTH. Models 3 and 4 control additionally for STOC VOL. Models 5 and 6 further add to Models 3 and 4 STOCKRET. Models 7 and 8 control additionally for TSPREAD, and for BBBSPREAD. Models and 10 control additionally for LIBOR. Models 11 and 12 use the NBER identification of peaks and troughs to identify the recessions in the Americ economy as opposed to the Stock–Watson Experimental Coincident Recession Index, which is used in all of the other models. Based on the new criteri REC equals one for the loans taken out between August 1990 and March 1991, and between April 2001 and November 2001. FIRM CONTROLS a LOAN CONTROLS coincide with the set of firm controls and loan controls, respectively, used in Table V. See Table I for variable definitions. Includ in the regressions but not shown in the table are also dummy variables for the issuer's sector of activity as defined by one-digit SIC code. Models a estimated with robust standard errors and clustered by firm to correct for correlation across observations of a given firm. Values in parentheses a p-values.

Variables	1	2	3	4	5	6	7	8	9	10	11	12
CONSTANT	635.821	636.958	513.911	512.543	509.745	508.020	584.911	585.192	632.787	634.598	631.866	633.21
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.00
MRPBOND	−37.385	−47.389	−31.726	−39.120	−31.873	−39.631	−34.069	−45.143	−38.339	−48.971	−41.195	−51.16
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.00
MRPBONDblg		33.319		28.213		29.499		36.216		34.871		34.88
		(0.000)		(0.001)		(0.001)		(0.000)		(0.000)		(0.00
MRPBONDnrt		35.838		24.049		24.281		38.045		35.651		32.59
		(0.008)		(0.033)		(0.031)		(0.004)		(0.008)		(0.00
REC	22.558	21.859	17.724	17.066	18.218	17.565	12.572	11.660	22.494	21.785	14.613	13.80
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)	(0.004)	(0.000)	(0.000)	(0.003)	(0.00
REC · MRPBOND	−20.882	−18.651	−20.840	−18.493	−20.589	−17.826	−20.004	−15.235	−21.092	−17.847	−21.317	−18.80
	(0.000)	(0.003)	(0.000)	(0.003)	(0.000)	(0.005)	(0.001)	(0.019)	(0.000)	(0.005)	(0.002)	(0.00
REC · MRPBONDblg		12.658		10.796		9.566		5.531		9.845		15.77
		(0.245)		(0.370)		(0.428)		(0.614)		(0.374)		(0.30

(*continue*

Table IX, part 1: robustness checks

Table IX—*Continued*

Variables	1	2	3	4	5	6	7	8	9	10	11	12
REC · MRPBONDnrt		−9.948		−11.482		−12.595		−15.968		10.398		12.80
		(0.713)		(0.619)		(0.585)		(0.561)		(0.701)		(0.73
SALESGROWTH	0.001	0.002										
	(0.837)	(0.666)										
STOCKVOL			9.371	9.389	9.694	9.731						
			(0.000)	(0.000)	(0.000)	(0.000)						
STOCKRET					−0.057	−0.060						
					(0.090)	(0.072)						
TSPREAD							11.420	11.432				
							(0.000)	(0.000)				
BBBSPREAD							41.981	42.794				
							(0.000)	(0.000)				
LIBOR									−1.235	−1.429		
									(0.131)	(0.078)		
FIRM CONTROLS	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN
LOAN CONTROLS	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN
Adjusted-R^2	0.4645	0.4671	0.4950	0.4969	0.4960	0.4980	0.4682	0.4709	0.4628	0.4655	0.4603	0.46
# Observations	13,846	13,846	10,201	10,201	10,201	10,201	13,846	13,846	13,846	13,846	13,846	13,8

Table IX, part 2: robustness checks

Table X

Loan Spreads for Bank-Dependent and Nondependent Borrowers: Robustness Checks: Risk in Recessions, Small Firms and Lending Relationships

The dependent variable is LOANSPREAD. Models 1 and 2 control for the interaction of REC with LEVERAGE. Models 3 and 4 are estimated on t sample of loans taken out by larger firms (excludes from the original sample all loans made taken out by the smaller firms defined by the first quart of sales). Models 5 through 8 control for our two alternative measures of firms' lending relationships: RLENDINGnarrow and RLENDINGbroa respectively. Models 9 through 12 investigate the effects of lending relationships in recessions by adding to Models 5 through 8 the interaction of RE with our two alternative measures of firms' lending relationships. FIRM CONTROLS and LOAN CONTROLS coincide with the set of firm contro and loan controls, respectively, used in Table V. See Table I for variable definitions. Included in the regressions but not shown in the table are al dummy variables for the issuer's sector of activity as defined by one-digit SIC code. Models are estimated with robust standard errors and cluster by firm to correct for correlation across observations of a given firm. Values in parentheses are *p*-values.

Variables	1	2	3	4	5	6	7	8	9	10	11	12
CONSTANT	629.954	630.880	564.388	562.371	629.780	630.791	632.345	633.338	629.665	630.810	632.489	633.73
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.00
MRPBOND	−37.922	−48.945	−39.687	−49.417	−38.158	−48.772	−38.216	−48.886	−37.509	−47.926	−37.110	−47.37
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.00
MRPBONDblg		35.875		32.827		34.729		34.882		34.424		34.21
		(0.000)		(0.000)		(0.000)		(0.000)		(0.000)		(0.00
MRPBONDnrt		35.771		28.680		35.678		35.609		35.399		11.64
		(0.007)		(0.028)		(0.007)		(0.007)		(0.008)		(0.00
REC	11.023	9.859	28.049	27.296	22.509	21.807	22.537	21.833	16.863	16.115	12.474	11.64
	(0.064)	(0.102)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.00
REC · MRPBOND	−21.171	−16.281	−24.507	−22.575	−20.402	−16.650	−20.638	−16.760	−22.605	−19.454	−24.897	−22.36
	(0.000)	(0.011)	(0.000)	(0.001)	(0.001)	(0.010)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)	(0.00
REC · MRPBONDblg		3.433		12.261		8.417		7.922		9.964		11.94
		(0.763)		(0.282)		(0.443)		(0.470)		(0.363)		(0.27

(continue

Table X, part 1: relationship lending robustness

Table X—*Continued*

Variables	1	2	3	4	5	6	7	8	9	10	11	12
REC · MRPBONDnrt		−12.506		−1.968		−12.584		−12.746		−7.601		−4.70
		(0.636)		(0.945)		(0.644)		(0.641)		(0.779)		(0.86
RLENDINGnarrow					−5.164	−5.260			−8.877	−8.996		
					(0.035)	(0.031)			(0.001)	(0.001)		
RLENDINGbroad							−9.772	−9.843			−14.972	−15.09
							(0.000)	(0.000)			(0.000)	(0.00
REC · RLENDINGnarrow									18.195	18.340		
									(0.003)	(0.003)		
REC · RLENDINGbroad											25.479	25.79
											(0.000)	(0.00
LEVERAGE	30.124	27.708	24.628	22.279	35.952	33.792	36.848	34.878	36.004	33.857	36.931	34.77
	(0.002)	(0.004)	(0.002)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.00
REC · LEVERAGE	35.606	37.004										
	(0.043)	(0.038)										
FIRM CONTROLS	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN
LOAN CONTROLS	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN
Adjusted-R^2	0.4631	0.4657	0.4621	0.4654	0.4629	0.4655	0.4636	0.4662	0.4635	0.4661	0.4649	0.46
# Observations	13,846	13,846	10,386	10,386	13,846	13,846	13,846	13,846	13,846	13,846	13,846	13,84

Table X, part 2: relationship lending robustness

7.8 Table XI: additional robustness

Read:

- Table XI reports further robustness tests, including reduced samples and alternative controls.
- The core signs and economic interpretation continue to support the paper's three main hypotheses.

Economic message: The broad robustness battery supports the conclusion that bank dependence is costly in recessions because banks can exploit informational advantages.

Loan Spreads for Bank-Dependent and Non-Bank-Dependent Borrowers: Robustness Checks: Sample Definition, Firm Fixed Effects, and Simultaneous Clustering by Firm and Time

The dependent variable is LOANSPREAD. Models 1 through 6 are clustered by firm to correct for correlation across observations of a given firm. Models and 8 are estimated with firm fixed effects. Models 1 and 2 are estimated on the sample made of one facility per loan deal. In the case of deals with multip facilities, we select the facility for this sample randomly. Models 3 and 4 are estimated on the sample of loans taken out during the 1990–2002 tin period (excludes from the original sample loans taken out in the 1987–1989 time period). Models 5 and 6 are estimated on the sample of nonrenegotiat loans (excludes from the original sample loan renegotiations). Models 7 and 8 are estimated with our original sample with robust standard errors ar firm fixed effects. Models 9 and 10 are also estimated with our original sample, but are clustered both by firm and time (to correct for correlation acro observations within each quarter). FIRM CONTROLS and LOAN CONTROLS coincide with the set of firm controls and loan controls, respectively, us in Table V. See Table 1 for variable definitions. Models estimated with robust standard errors. Included in the regressions but not shown in the table a also dummy variables for the issuer's sector of activity as defined by one-digit SIC code. Values in parentheses are *p*-values.

Variables	1	2	3	4	5	6	7	8	9	10
CONSTANT	649.596	651.060	635.660	635.739	623.770	624.740	400.542	400.555	628.031	629.02
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.00
MRPBOND	−38.111	−46.261	−30.657	−40.398	−31.959	−41.684	−42.737	−42.823	−38.073	−48.64
	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.002)	(0.000)	(0.00
MRPBONDblg		33.297		31.674		32.704		−2.256		34.67
		(0.000)		(0.000)		(0.000)		(0.861)		(0.00
MRPBONDnrt		39.209		23.621		40.433		14.821		35.48
		(0.001)		(0.026)		(0.008)		(0.501)		(0.02
REC	20.350	19.704	24.328	23.619	22.395	21.773	26.180	26.209	22.492	21.79
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.00
REC · MRPBOND	−18.634	−15.600	−19.659	−15.959	−20.686	−18.902	−18.434	−14.064	−20.222	−16.48
	(0.001)	(0.006)	(0.001)	(0.013)	(0.001)	(0.005)	(0.008)	(0.056)	(0.004)	(0.03
REC · MRPBONDblg		7.372		5.5678		17.832		−21.419		8.32
		(0.522)		(0.621)		(0.122)		(0.113)		(0.58
REC · MRPBONDnrt		−8.554		−2.561		−14.521		3.579		−11.70
		(0.783)		(0.927)		(0.570)		(0.924)		(0.70
FIRM CONTROLS	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN
LOAN CONTROLS	IN	IN	IN	IN	IN	IN	IN	IN	IN	IN
Adjusted-R^2	0.4688	0.4713	0.4769	0.4790	0.4700	0.4726	0.2788	0.2759	0.4626	0.46
# Observations	9,857	9,857	12,505	12,505	11,887	11,887	13,846	13,846	13,846	13,84

Table XI: additional robustness tests

8 Detailed result synthesis

8.1 Bond access effect

Across specifications, public bond market access is associated with lower loan spreads. This is consistent with the idea that outside financing options improve the borrower's bargaining position in bank loan negotiations.

8.2 Recession effect

Loan spreads rise in recessions, consistent with both higher borrower default risk and higher economy-wide credit risk premia. The key issue is whether this increase is larger for bank-dependent borrowers.

8.3 Recession-by-access effect

The negative recession-by-access interaction implies that bond-access firms are protected from part of the recession increase in loan spreads. This is the strongest evidence for the informational hold-up mechanism.

8.4 Risk interaction effect

The access discount is larger for riskier firms. This is precisely where adverse selection should be strongest and where a relationship bank's private information should be most valuable.

8.5 Relationship-lending interpretation

Bank relationships are valuable because they support monitoring and credit provision. But the same relationship can become costly when it gives the bank private information and the borrower lacks outside options.

9 Short summary

- The paper tests whether banks exploit informational monopolies over borrowers.
- The theory predicts that bank hold-up is stronger when borrowers are risky.
- Recessions are used as states in which borrower risk and adverse-selection concerns rise.
- The empirical comparison is between bank-dependent borrowers and borrowers with public bond market access.
- Firms with public bond market access pay lower bank loan spreads.
- Loan spreads rise in recessions, but much less for firms with public bond market access.
- The most recent public bond measure is the strongest access proxy.
- The effect remains after controlling for firm risk, loan characteristics, and sector effects.
- The effect is stronger for riskier firms.
- IV and robustness tests support the baseline interpretation.
- The conclusion is that bank relationships have cyclical hold-up costs: dependence on banks is especially expensive during recessions.

10 Conclusion

10.1 Core conclusion

Santos and Winton show that bank-dependent borrowers pay higher loan spreads, and that this disadvantage becomes larger in recessions. Firms with access to public debt markets pay lower loan spreads and are less exposed to recession-induced spread increases.

10.2 Economic intuition

Banks monitor borrowers and learn private information. This creates value by enabling credit provision, but it also creates market power. When the borrower is risky and outside lenders fear adverse selection, the relationship bank can extract rents. Recessions make this mechanism stronger because borrower risk and information problems rise.

10.3 Takeaway

The paper's main lesson is that relationship banking has both benefits and costs. Public bond market access functions as an outside option that limits banks' informational monopoly power. The cost of losing that outside option is most visible in bad times.